

1. Background

The impact of solar mini-grids can affect positively to socioeconomics. This letter specifically focus on improvements in education using solar mini-grids. This analysis examines the effect of solar mini-grid on the improvement of education in Kenya and Nigeria by analyzing school performance change as the primary dependent variable.

2. Data cleaning

2.1 We exclude all text variable but the school performance change reason and make the following change to variables:

For columns with 2 choices (Such as yes,no) , converting into binary variable

For columns with 3 choices (Such as Yes,increased/No same/Yes, decreased), converting into ordinal variable (like I/N/D).

For numerical variable (such as energy usage) keep it as its original value

For column with number scope (such as 100-200, 200-500,500-1000), choose the ceiling value.

2.2 Then we do a principal component analysis to combine some variables measured similar aspects. And we get 26 components finally.

3. Model selection

We use the school_performance_change as the dependent variable.

We use the logistic regression(ridge), logistic regression (Lasso), random forest, gradient boosting and SVM 5 model approaches to help us to filter the key components that make significant impact to the dependent variables.

Through the 5 above model, we filter out 8 variables that make significant impact to the dependent variables.

4. statistical test

Then based on the different variable types, we did statistic tests to check if these variables show significant difference within different dependent variables group.

We exclude the variables that failed in statistic tests and focus on the variables passed the tests

5. selected feature analysis

We analyze the power and direction of the filtered variables, explore how these variables set up connection with school performance change in reality, and how they were impacted by mini-grid projects.

Then we summarize our findings to analyze how mini-grid impacted the education of the local communities directly and indirectly.